ce Assignment • 15 min

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

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1. What type of data are Convolutional Neural Networks (CNNs) primarily designed to process?

- ○ Sequential data, such as text or time series
- ○ Tabular data with structured features
- ◉ Grid-like data, such as images and video
- ○ Audio data

Correct
This is Correct. CNNs excel at handling grid-like data due to their ability to capture spatial relationships within the data using convolutional layers. For more information, refer to the video "Understanding the Convolutional Neural Networks".

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2. What is the primary purpose of Pooling Layers in a CNN?

- ○ To increase the spatial dimensions of the data
- ○ To introduce non-linearity into the model
- ◉ To reduce the spatial dimensions of the data by downsampling
- ○ To generate the final output predictions

Correct
This is Correct. Pooling layers downsample the feature maps, reducing their spatial dimensions and making the network more computationally efficient and robust to small shifts in the input. For more information, refer to the video "Understanding the Convolutional Neural Networks".

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3. What is the core concept behind transfer learning?

- ○ Training a model from scratch on a small dataset.
- ◉ Leveraging knowledge from a pre-trained model on a new but related task.
- ○ Creating a completely new neural network architecture for every task.
- ○ Only using labeled data for training.

Correct
This is Correct. Transfer learning involves taking a model that has been trained on a large dataset and

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4. In which scenario is transfer learning most likely to be beneficial?

- ○ You have abundant labeled data for your specific task.
- ○ The pre-trained model was trained on a task completely unrelated to your target task.
- ○ You have ample computational resources and a large dataset for your new task.
- ◉ You have a small dataset for your specific task and limited computational resources.

Correct
This is Correct. Transfer learning is particularly advantageous when you have a small dataset and limited computational power, as it allows you to leverage the knowledge from a pre-trained model. For more information, refer to the video "Transfer Learning Techniques".